Eigenvalues and inverse problems

A condition number of two for a sign matrix in every dimension

Difficulty: challenging **Importance:** interesting to the community

Status: Partially resolved

Independently reviewed partial result - 2026-09-11

Theorem 1 constructs explicit real sign matrices of order p^2 for every odd prime $p \geq 13$, with condition number at most $(p+2+3\sqrt{p})/(p-2-3\sqrt{p})$. For primes $p \geq 361$ this is at most $210/151 < \sqrt{2}$. This answers the cited paper's separate Problem 13, but does not prove the repository's bound in every dimension.

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Rating rationale: Challenging reflects a sharp universal constant whose remaining obstruction is a large finite set of dimensions; community impact is the construction of well-conditioned sign matrices.

Problem statement

For $n \geq 1$, define

$$h(n) = \min_{A \in \{-1,1\}^{n \times n}} \kappa_2(A), \qquad \kappa_2(A) = \frac{\sigma_{\max}(A)}{\sigma_{\min}(A)},$$

with $\kappa_2(A) = +\infty$ for singular A. Prove or disprove

$$\sup_{n \geq 1} h(n) = 2.$$

Since $h(3) = 2$, the unresolved claim is the upper bound in every dimension. No symmetry or circulant structure is imposed. The target refines the known existence of uniformly well-conditioned sign matrices, which is useful for constructing approximately isometric linear maps and well-conditioned bases.

References

Alexeev, Jasper, and Mixon, *Asymptotically optimal approximate Hadamard matrices*, §6, Problem 12. Dong and Rudelson, *Approximately Hadamard matrices and Riesz bases in random frames*, introduction; *IMRN* (2024), 2044–2065.

Earlier status check — 2026-09-08

Searches for *approximate Hadamard sup 2 conjecture*, *2511.14653 2026 condition number*, and *approximate Hadamard supremum* found no proof of the sharp constant. The fact that $h(n) \to 1$ is already known and is not the problem counted here.

Audit update — 2026-09-10

Rechecked Alexeev–Jasper–Mixon, §6, Problem 12. Their theorem $h(n) \to 1$ implies the requested upper bound for every sufficiently large dimension, leaving finitely many dimensions unresolved; this is substantive partial resolution of the displayed universal statement. The numerical cutoff quoted there assumes the Hadamard conjecture and is not an unconditional completion. Supremum-two searches found no later full proof.